DEVELOPER API REFERENCE

SAAPT 2.0 Backend Action-Based API Collection

Clear, Explicit Action Endpoints (add-student, delete-student, list-students, etc.)

BASE URL (LOCAL)
http://127.0.0.1:8000 / http://127.0.0.1:8001

API STYLE
Action-Oriented (add-student, edit-student, delete-student)

AUTHENTICATION
JSON Web Token (JWT Bearer Token)

DATABASE
PostgreSQL (Supabase) / SQLite

CONTENT TYPES
application/json & multipart/form-data

RELEASE VERSION
Version 2.0 • Updated October 2026

1. Student Management (Action-Based APIs)

Base: /api/

POST /api/add-student/

Bearer Token Required | multipart/form-data

Add Student API: Creates a new student admission record with personal details, parent information, and uploaded certificates.

FORM DATA REQUEST PARAMETERS

Field	Type	Required	Description
full_name	string	Required	Full legal name of the student.
date_of_birth	date (YYYY-MM-DD)	Required	Student date of birth.
gender	string	Required	Male, Female, or Other.
academic_year	integer (ID)	Required	Academic year ID (e.g. 1).
class_section	integer (ID)	Optional	Assigned class section ID.
roll_number	string	Optional	Class roll number.
blood_group	string	Optional	e.g. O+, A+, B+.
father_name	string	Optional	Parent / Father name.
mother_name	string	Optional	Mother name.
mobile_number	string	Optional	Primary contact phone number.
parent_email	string	Optional	Parent email address.
address	string	Optional	Full residential address.
student_photo	file	Optional	Passport size photograph.

RESPONSE (201 CREATED)


```
{
  "id": 102,
  "admission_number": "SAAPT-2026-042",
  "full_name": "Ananya Patel",
  "status": "active",
  "date_of_birth": "2015-09-20",
  "gender": "Female",
  "blood_group": "B+",
  "academic_year": 1,
  "class_section": 2,
  "roll_number": "08"
}
```

GET /api/list-students/

Bearer Token Required | application/json

List Students API: Retrieves the student directory table with pagination, name/admission search, and class/section filters.

QUERY PARAMETERS

Parameter	Type	Description
search	string	Search student name or admission number.
class	string/int	Filter by class name or class ID (e.g. Class 6 or 1).
section	string/int	Filter by section letter or section ID (e.g. A).
status	string	active, transferred, graduated, inactive.
academic_year	string/int	Filter by academic year (e.g. 2025-2026).
page	integer	Page number (default 10 items per page).

RESPONSE (200 OK)


```
{
  "count": 245,
  "next": "http://127.0.0.1:8000/api/list-students/?page=2",
  "previous": null,
  "results": [
    {
      "id": 101,
      "admission_number": "SAAPT-2025-001",
      "full_name": "Aarav Sharma",
      "class_display": "Class 6 - A",
      "roll_number": "12",
      "parent_name": "Rajesh Sharma",
      "parent_mobile": "+919876500001",
      "status": "active",
      "admission_date": "2025-06-01"
    }
  ]
}
```

GET /api/view-student/<id>/

Bearer Token Required | application/json

View Student Profile API: Returns complete student profile with parent info, documents, KPIs, and marksheet records.

RESPONSE (200 OK)

```
{
  "id": 101,
  "admission_number": "SAAPT-2025-001",
  "full_name": "Aarav Sharma",
  "date_of_birth": "2014-05-12",
  "gender": "Male",
  "blood_group": "O+",
  "class_display": "Class 6 - A",
  "roll_number": "12",
  "status": "active",
  "overall_grade": "A+",
  "attendance_percentage": 94.50,
  "parent_guardian": {
    "father_name": "Rajesh Sharma",
    "mother_name": "Priya Sharma",
    "mobile_number": "+919876500001",
    "email": "rajesh.sharma@example.com"
  }
}
```

PUT PATCH /api/edit-student/<id>/

Bearer Token Required | multipart/form-data

Edit Student API: Updates an existing student's record and parent information.

REQUEST BODY EXAMPLE (PATCH)

```
{
  "roll_number": "15",
  "status": "active",
  "mobile_number": "+919876599999"
}
```

RESPONSE (200 OK)

```
{
  "id": 101,
  "full_name": "Aarav Sharma",
  "roll_number": "15",
  "status": "active"
}
```

DELETE /api/delete-student/<id>/

Bearer Token Required

Delete Student API: Permanently removes a student's profile and associated academic records from the database.

RESPONSE

```
204 No Content (Empty Response Body)
```

GET /api/student-stats/

Bearer Token Required application/json

Student Stats API: Returns dashboard KPI metrics including total student counts, active/transferred counts, and per-class enrollment breakdowns.

RESPONSE (200 OK)


```
{
  "total_students": 450,
  "active_students": 420,
  "transferred_students": 12,
  "graduated_students": 15,
  "new_admissions_this_month": 28,
  "class_breakdown": [
    {
      "id": 1,
      "school_class__name": "Class 6",
      "section__name": "A",
      "max_strength": 40,
      "active_count": 38
    }
  ]
}
```

POST /api/bulk-allocate-students/

Bearer Token Required | application/json

Bulk Allocate Students API: Atomically reallocates selected students to a new section, ensuring classroom capacity is not exceeded.

REQUEST BODY

```
{
  "student_ids": [101, 102, 105],
  "target_class_section_id": 3
}
```

RESPONSE (200 OK)

```
{
  "message": "Successfully moved 3 students to Class 7 - B.",
  "target_class_section": {
    "id": 3,
    "display_name": "Class 7 - B",
    "current_strength": 35,
    "max_strength": 40
  }
}
```

GET /api/export-students/ & /api/student-id-card/<id>/

Bearer Token Required

Export Students: Generates CSV spreadsheet download.
Student ID Card: Returns printable student badge information.

STUDENT ID CARD RESPONSE (200 OK)

```
{
  "school_name": "SAAPT International Public School",
  "student_name": "Aarav Sharma",
  "admission_number": "SAAPT-2025-001",
  "class_section": "Class 6 - A",
  "roll_number": "12",
  "blood_group": "O+",
  "emergency_contact": "+919876500001",
  "photo_url": "http://127.0.0.1:8000/media/students/photos/aarav.jpg"
}
```

2. Class & Section Allocation (Action-Based APIs)

Base: /api/

GET `/api/list-class-sections/`

Bearer Token Required | application/json

List Class Sections API: Lists all allocated sections, assigned teachers, room numbers, and current vs. max student capacity.

RESPONSE (200 OK)


```
[
  {
    "id": 1,
    "academic_year": 1,
    "academic_year_name": "2025-2026",
    "school_class": 1,
    "class_name": "Class 6",
    "section": 1,
    "section_name": "A",
    "room_number": "Room 101",
    "class_teacher_name": "Mrs. Anjali Rao",
    "max_strength": 40,
    "current_strength": 38,
    "display_name": "Class 6 - A"
  }
]
```

POST /api/add-class-section/

Bearer Token Required | application/json

Add Class Section API: Configures a new class and section allocation record.

REQUEST BODY

```
{
  "academic_year": 1,
  "school_class": 2,
  "section": 1,
  "room_number": "Room 204",
  "class_teacher_name": "Mr. Vikram Sen",
  "max_strength": 45
}
```

RESPONSE (201 CREATED)

```
{
  "id": 5,
  "class_name": "Class 7",
  "section_name": "A",
  "display_name": "Class 7 - A",
  "current_strength": 0,
  "max_strength": 45
}
```

PUT DELETE /api/edit-class-section/<id>/ & /api/delete-class-section/<id>/

Bearer Token Required

Edit Class Section: Modifies room number, teacher, or capacity.
Delete Class Section: Removes an empty class section allocation.

STUDENTS IN SECTION (GET /API/CLASS-SECTION-STUDENTS/<ID>/)

```
[
  {
    "id": 101,
    "admission_number": "SAAPT-2025-001",
    "full_name": "Aarav Sharma",
    "roll_number": "12",
    "status": "active"
  }
]
```

3. Authentication & Password Security

Base: /api/auth/

POST /api/auth/forgot-password/

No Auth Required | application/json

Forgot Password API (Step 1 - Ask Mail ID & Send OTP): Asks user for their registered email address and sends a secure 6-digit OTP code (valid for 10 minutes).

REQUEST BODY

```
{
  "email": "user@saapt.com"
}
```

RESPONSE (200 OK)

```
{
  "message": "OTP has been sent to your registered email address.",
  "email": "user@saapt.com"
}
```

POST /api/auth/verify-otp/

No Auth Required | application/json

Verify OTP API (Step 2 - Verify Code): Validates the 6-digit code entered by the user. If correct, returns verification confirmation and reset token for step 3.

REQUEST BODY

```
{
  "email": "user@saapt.com",
  "otp": "582914"
}
```

RESPONSE (200 OK)

```
{
  "message": "OTP verified successfully. You can now enter your new password.",
  "email": "user@saapt.com",
  "reset_token": "u9mP0q-vA7Xz1bKr_8xL92aBcD4eFgHiJkLmNoPqRs"
}
```

POST /api/auth/reset-password/

No Auth Required | application/json

Reset Password API (Step 3 - Set New Password & Confirm): Takes the new password and confirmation along with the verified OTP or reset token to update the password.

REQUEST BODY

```
{
  "email": "user@saapt.com",
  "reset_token": "u9mP0q-vA7Xz1bKr_8xL92aBcD4eFgHiJkLmNoPqRs",
  "new_password": "NewSecurePass@2026",
  "confirm_password": "NewSecurePass@2026"
}
```

RESPONSE (200 OK)

```
{
  "message": "Password changed successfully. You can now log in with your new password."
}
```

POST /api/auth/login/ & /api/auth/register/

No Auth Required | application/json

Login API: Log in via username or email.
Register API: Staff user registration.

LOGIN REQUEST BODY

```
{
  "username_or_email": "user@saapt.com",
  "password": "NewSecurePass@2026"
}
```

LOGIN RESPONSE (200 OK)

```
{
  "message": "Login successful.",
  "user": {
    "id": 14,
    "username": "sarah_admin",
    "email": "user@saapt.com"
  },
  "tokens": {
    "access": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "refresh": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
  }
}
```

4. Master Lookups & Dropdowns (Action-Based APIs)

Base: /api/

GET /api/get-classes/ & /api/get-sections/ & /api/get-academic-years/

Bearer Token Required

Dropdown master data endpoints used for populating dropdown selections in forms and filter controls.

EXAMPLE: GET /API/GET-CLASSES/ RESPONSE (200 OK)

```
[
  { "id": 1, "name": "Class 6", "code": "CLS-6", "numeric_order": 6 },
  { "id": 2, "name": "Class 7", "code": "CLS-7", "numeric_order": 7 }
]
```

EXAMPLE: GET /API/GET-SECTIONS/ RESPONSE (200 OK)

```
[
  { "id": 1, "name": "A" },
  { "id": 2, "name": "B" }
]
```

EXAMPLE: GET /API/GET-ACADEMIC-YEARS/ RESPONSE (200 OK)

```
[
  {
    "id": 1,
    "year_name": "2025-2026",
    "start_date": "2025-06-01",
    "end_date": "2026-04-30",
    "is_current": true
  }
]
```